

- 4 A student sets up the apparatus illustrated in Fig. 4.1 in order to investigate the oscillations of a 100 g metal cube suspended on a string.

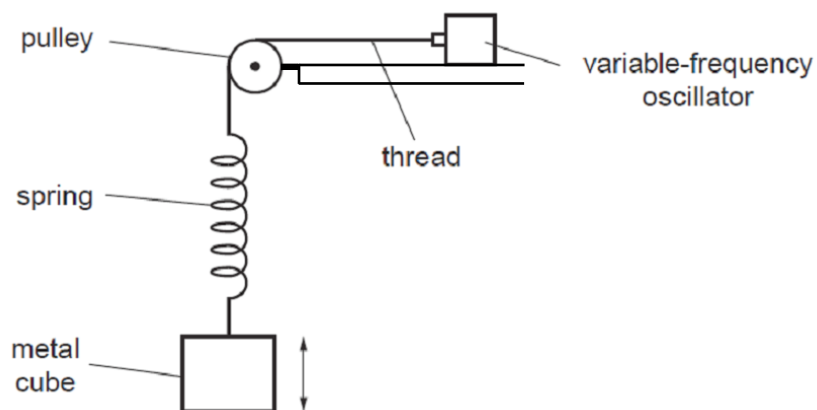


Fig. 4.1

Before switching on the variable-frequency oscillator, the metal cube was displaced by pulling it downwards and made to oscillate freely. A graph of its velocity against displacement is shown in Fig. 4.2.

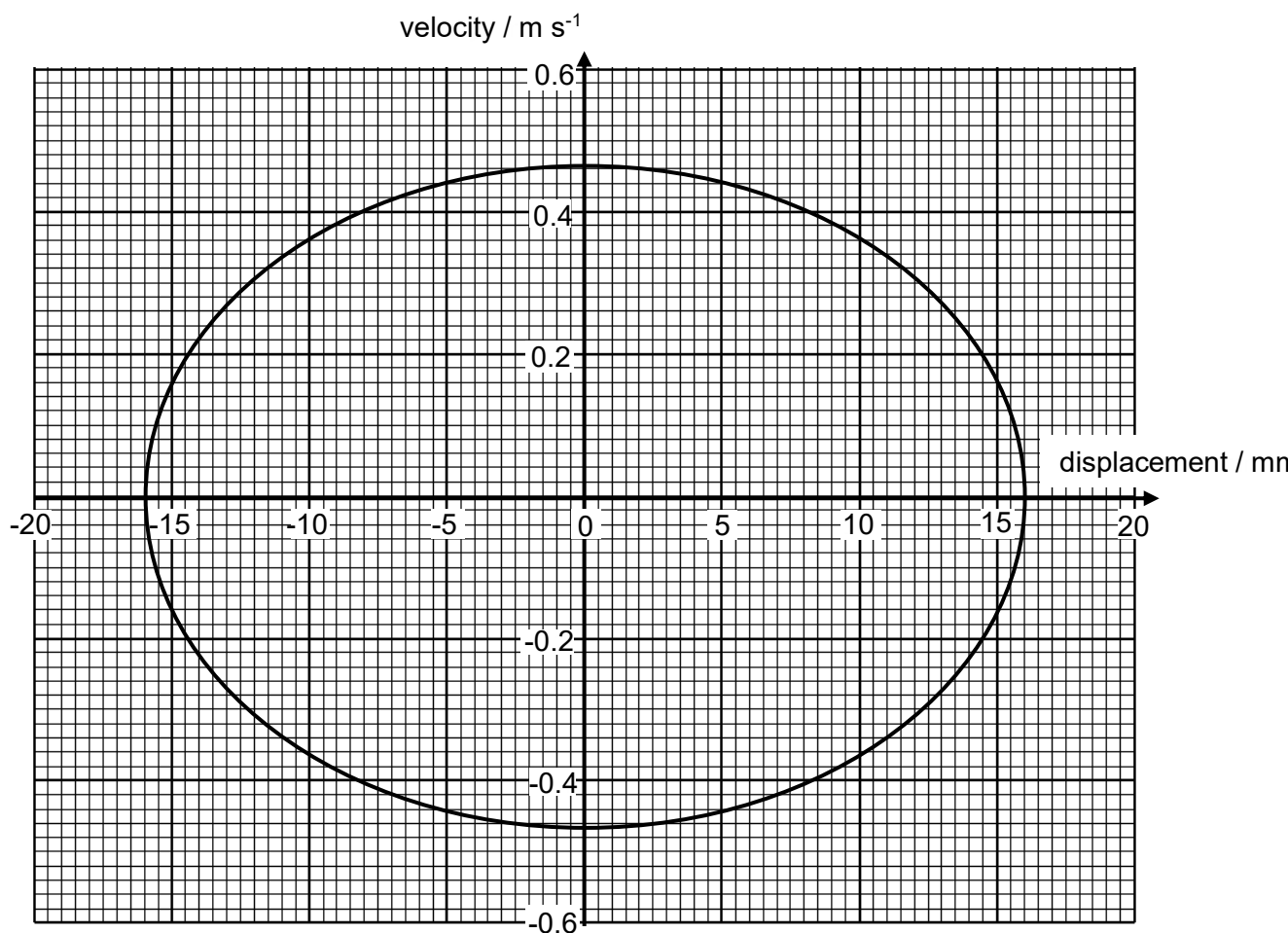


Fig. 4.2

- (a) Using Fig. 4.2, show that the natural frequency of the metal cube is 4.6 Hz.

From graph, $v_0 = 0.46 \text{ m s}^{-1}$ and $x_0 = 16.0 \text{ mm}$.

$$v_0 = \omega x_0$$

$$v_0 = 2\pi f x_0$$

$$0.46 = 2\pi f (16.0 \times 10^{-3})$$

$$f = 4.6 \text{ Hz} \quad [\text{shown}]$$

[2]

- (b) Hence, or otherwise, determine the maximum resultant force acting on the metal cube.

$$a_0 = \omega^2 x_0 = (2\pi f)^2 x_0 = [2\pi(4.6)]^2 (16.0 \times 10^{-3}) = 13.37 \text{ m s}^{-2}$$

By Newton's 2nd Law of motion,

$$\Sigma F = ma$$

$$\Sigma F_0 = ma_0 = 0.100 (13.37) = 1.3 \text{ N}$$

maximum resultant force = N [2]

- (c) The variable-frequency oscillator is now switched on so that the metal cube can undergo forced oscillation.

- (i) State what is meant by *forced oscillation*.

..... Forced oscillation is an oscillatory system which is driven by an external periodic force.

[1]

- (ii) On Fig. 4.3, sketch the variation with frequency of the amplitude of the oscillations of the metal cube between 2 Hz to 8 Hz.

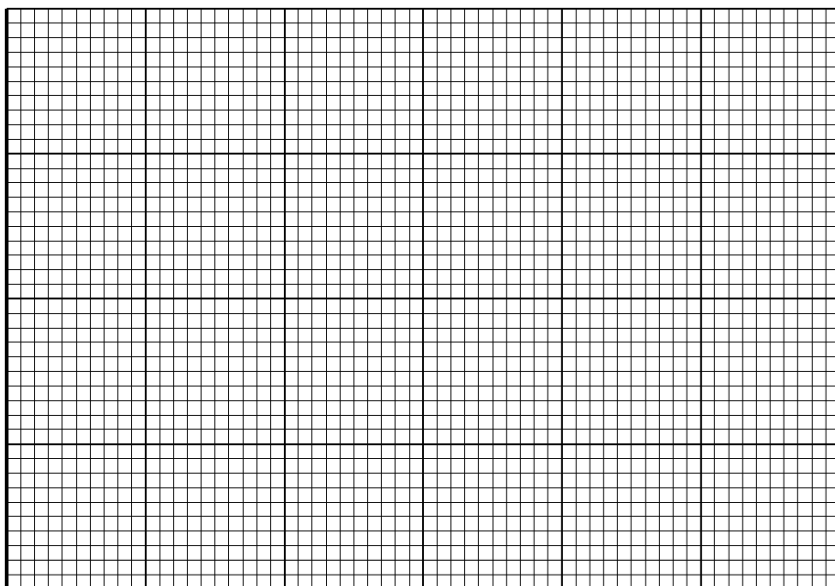


Fig. 4.3

A resonance graph with the correct shape with peak at 4.6 Hz.

[2]

[Total: 7]